

GitHub 101: Saving, Sharing, and Undoing Your Work

THE IDEA IN ONE LINE

A repository is a project folder that remembers every version, and why each one changed.

FOUR WORDS

WORD	WHAT IT IS	WHEN YOU USE IT
Repository	A project folder with its full history	One per project
Commit	A saved version with a one-line reason	Whenever you reach a state worth returning to
Branch	A parallel draft of the whole project	To try something without touching the main version
Pull request	A proposal to fold a branch back in, with every change shown	Before anything joins main, even your own work

Git is the program that keeps the history. GitHub is the website that hosts it.

YOUR FIRST REPOSITORY, IN THE BROWSER

- Create:** + menu, New repository. Private. Tick "Add a README file."
- Add:** Add file, Create new file, `prompts.md`. Commit changes, with a message that says why.
- Branch:** edit any file (the pencil), choose "Create a new branch," Propose changes.
- Review:** Create pull request, read the diff (red out, green in), Merge.
- Edit anywhere:** press `.` on any repository to open it in a full editor.

THE DAILY LOOP

Pull before you start. Edit. Commit with a reason. Push when you stop. A good message: Drop 2019 rows counted twice in the merge. A useless one: `update`.

WHEN IT BREAKS

Nothing you committed is lost; any version can be restored. Paste the error into the model:

Explain what happened in plain language and give me the smallest safe fix. No force-push, no reset --hard.

QUICK WINS

- Private by default. Make a repository public on purpose.
- Write the README first, for yourself in six months.
- Plain text shows its changes (`.md`, `.tex`, `.csv`, `.py`). Word and PDF files are stored, but you can't see what changed.
- Tag a release when you send a draft to your committee. You can always return to exactly what they read.

NEVER PUSH

- Passwords, API keys, tokens. Keep them in `.env` and list it in `.gitignore`.
- Sensitive, proprietary, or licensed data, or RAND work nobody has cleared.
- Copyrighted readings. Files over 100 MB (GitHub refuses them).
- Deleting a file later does not remove it from the history. If a key leaks, revoke it first.

WHERE THE MODEL FITS

- Agents edit files; the history turns each edit into a diff you read before you keep it. (Sessions 3 and 4)
- Skill docs travel as repositories. (Session 5)
- A `Co-Authored-By` line in a commit records AI help. (Session 1)
- Verified students get Copilot free through GitHub Education.

GO DEEPER

- GitHub Skills, short interactive courses: skills.github.com
- GitHub Docs, Get started: docs.github.com/en/get-started
- GitHub Desktop, Git without the terminal: desktop.github.com
- Software Carpentry, *Version Control with Git*: swcarpentry.github.io/git-novice
- Picking a license: choosealicense.com
- Fixes for common mistakes: dangitgit.com